Do Pakistanis Hold Militant Groups in Low Regard?*

Gaurav Sood†

Abstract

Blair et al. (2013) find that endorsements by militant groups slightly lower Pakistanis' support for policies and conclude that Pakistanis hold these groups in "low regard." That conclusion reads the average as the typical respondent's reaction. The same small negative average could come from widespread mild disapproval, from indifference, or from large reactions in both directions that nearly offset. The average cannot separate these readings, but the design can: each respondent answers four policy questions, so the control arm shows how a respondent's answers move together without endorsements. Under endorsement, answers to most pairs of policies become more closely related, and the share of respondents whose answers average the midpoint of the scale falls by 3.8 percentage points, with gains on both sides. Simulations built from control respondents show that opposition alone does not produce this pattern. The small average effect reflects offsetting reactions, not mild disapproval or indifference. How many respondents react favorably is not identified: the standard measurement model's estimate of 43% assumes that a respondent's reactions are independent across questions, which the data contradict.

1 Introduction

Do Pakistanis hold militant groups in low regard? Blair et al. (2013) use endorsement experiments to measure affect toward four militant groups. They find small negative average effects on stated policy support and interpret them as weakly negative average affect. That interpretation leaves a different question open: how widely are favorable or unfavorable attitudes held?

The small average effect reflects offsetting reactions, not widespread mild disapproval or indifference. Because each respondent answers four policy questions, the control arm supplies the counterfactual for how a respondent's answers move together, not only for each answer separately. Under endorsement, answers to five of the six policy pairs become more closely related, which reactions carried across questions produce and independent noise does not. Respondents whose four answers average the midpoint of the scale become 3.8 percentage points rarer, with gains on both sides. Endorsed arms simulated from the control respondents show that opposition alone, at each of three intensities that reproduce the average effect, leaves that share unchanged (Section 4). The argument assumes that how a respondent reacts is unrelated to where their answers would otherwise sit.

*https://github.com/finite-sample/endorsement.

†gsood07@gmail.com

The measurement model of Bullock et al. (2011) gives a 43% probability of a positive latent reaction to an endorsement, averaged equally across the four groups (95% credible interval 32%–53%), with mean reactions no larger than 0.08 ideal-point standard deviations and larger estimated dispersion. That model treats each reaction as independent across a respondent's questions, and the within-respondent patterns contradict it. The data show that substantial favorable and unfavorable reactions coexist; they do not pin down how many respondents react favorably.

2 Endorsement Experiments

Endorsement experiments randomize whether a policy question includes an endorsement by an actor of interest. Respondents in the control condition evaluate the policy alone; those in the treatment condition evaluate the policy with the endorsement. With valid assignment and observation, the difference identifies an effect of the cue on stated policy support.

Interpreting that effect as affect toward the endorser requires more. Respondents must understand the endorsement, and its effect must operate through their attitudes toward the group rather than through other information the cue conveys. Even then, the size of a response change can depend on baseline policy preferences and the response scale. The design has been used to study militant groups in Pakistan and combatants in Afghanistan (Bullock et al., 2011; Blair et al., 2013; Lyall et al., 2013; Blair et al., 2014).

A small negative ATE can reflect many small negative reactions, a few large ones, or positive and negative reactions that nearly cancel. Suppose a proportion p of respondents reacts favorably and the rest unfavorably, with average effects δ^+ and δ^- on the observed response. Then

$$\beta = p\delta^+ + (1-p)\delta^-. \tag{1}$$

One observed mean cannot determine three unknowns (Manski, 1990). Widespread mild disapproval (p near zero, small negative δ^-), indifference (δ^+ and δ^- near zero), and offsetting reactions (large δ^+ and δ^- of opposite sign) can all produce the same β. Nor does the sign of an effect, without a measurement assumption, classify a respondent as a supporter or opponent. Each respondent in this experiment answers several questions, and the way those answers move together separates the two readings (Section 4).

3 The Experiment and Data

Blair et al. (2013) surveyed 6,000 adults in Pakistan's four main provinces in 2009. Their experiment covered Kashmiri tanzeems, the Afghan Taliban, al-Qa'ida, and sectarian tanzeems. Respondents evaluated polio vaccination, Frontier Crimes Regulation reform, the Durand Line, and curriculum standardization on a five-point intensity scale. I rescale responses to $[0, 1]$; a mean on this scale is not a percentage of respondents supporting a policy.

I use the analysis file supplied with the `endorse` R package (Shiraito and Imai, 2022). It contains 5,212 respondents with one observed answer per policy, yielding 10,619 control responses and 10,229 endorsed responses. The pooled control mean is 0.736, the treatment mean is 0.723, and

their difference is −0.013 (respondent-clustered SE 0.0052). These are response-weighted means across the available policy items and endorsers.

The analysis is unweighted. Blair et al. use population and policy weights; their survey oversampled Balochistan and Khyber Pakhtunkhwa. Their unconditional Table 1 has 5,358 respondents and a control mean of 0.796. Differences from the present estimates can reflect sample selection and policy weighting as well as population weighting. The estimates below therefore describe the analysis sample, not nationally representative prevalence.

The package sample also retains assignment patterns excluded by the original replication code: 35 respondents have control and endorsed items across policies, and 94 receive the same endorser on multiple policies. Excluding these 129 respondents changes the pooled ATE from −0.01283 to −0.01272. This diagnostic leaves the sign and rounded magnitude intact. The Bayesian models retain the package sample; their sensitivity to these exclusions has not been estimated.

4 Within-Respondent Patterns

Each respondent answers all four policy questions. Endorsed respondents receive a different group on each question; control respondents receive none. Random assignment therefore gives the control arm's joint distribution of the four answers as the counterfactual for the endorsed arm. Two features of that joint distribution show whether respondents react: how a respondent's answers move together across policies, and the distribution of each respondent's mean answer. Neither requires a model. The 35 respondents with both control and endorsed items are excluded from this section; intervals are percentile bootstrap intervals over respondents.

If reactions are independent across a respondent's questions, as in the standard measurement model (Bullock et al., 2011), the correlation between any two policy answers is the same with and without an endorsement, apart from the effect of the mean shift on the discrete scale. Table 1 shows that the correlations change, and not in one direction. The two policies whose answers are most strongly related in the control arm, Frontier Crimes Regulation reform and the Durand Line, become less related under endorsement: the correlation falls from 0.66 to 0.54 (−0.12, 95% interval [−0.16, −0.07]). Every other pair becomes more related, by 0.07 to 0.18, with every interval excluding zero. A reaction that each respondent carries across all four endorsements raises every pairwise correlation; a cue that also loosens the link between an answer and the respondent's policy position lowers the correlation for the pair most tightly linked to that position. The pattern is consistent with both operating together. It is not consistent with reactions that are independent across questions, and it is not consistent with reactions that enter each question in proportion to that question's loading on the latent position, which would raise the FCR–Durand correlation most.

Table 2 compares the distribution of each respondent's mean answer across the four policies. Indifference predicts no change. In the control arm, 10.8% of respondents have answers that average exactly the midpoint of the scale; under endorsement 7.0% do, a change of −3.8 [−5.4, −2.3] percentage points. Mass appears on both sides of the midpoint: +2.1 [+1.0, +3.1] points one step below and +2.7 [+1.1, +4.3] points one step above, while the bin two steps above is unchanged (+0.1 [−1.7, +1.8]). The gain one step above the midpoint cannot have come from

Table 1: Correlations between policy answers, by arm

Policy pair	Control	Endorsed	Difference	95% interval
Polio – FCR	0.08	0.16	+0.08	[+0.02, +0.14]
Polio – Durand	0.02	0.11	+0.09	[+0.03, +0.14]
Polio – Curriculum	0.16	0.23	+0.07	[+0.02, +0.13]
FCR – Durand	0.66	0.54	−0.12	[−0.16, −0.07]
FCR – Curriculum	0.13	0.31	+0.18	[+0.13, +0.24]
Durand – Curriculum	0.14	0.25	+0.11	[+0.06, +0.17]

Pearson correlations between the rescaled answers to two policies, across respondents within each arm. Intervals are 95% percentile bootstrap intervals for the difference, resampling respondents within arms. Respondents with both control and endorsed items are excluded.

above, so some respondents at the midpoint moved up. The rest of the distribution drifts down: the lower tail gains +1.7 [+0.7, +2.7] points and the upper bins lose a little, although those changes are not distinguishable from zero. The share of respondents at the top of the scale on all four policies is 16.4% in the control arm and changes by −0.3 [−2.1, +1.8] points under endorsement. Favorable reactions among the 31.1% of respondents averaging 0.875 or higher are largely invisible by construction; unfavorable reactions anywhere are not.

Table 2: Distribution of each respondent's mean answer, by arm

Mean answer	Control	Endorsed	Difference (pp)	95% interval
≤ 0.375	2.9%	4.6%	+1.7	[+0.7, +2.7]
0.4375	2.9%	4.9%	+2.1	[+1.0, +3.1]
0.5	10.8%	7.0%	−3.8	[−5.4, −2.3]
0.5625	7.7%	10.4%	+2.7	[+1.1, +4.3]
0.625	12.3%	12.4%	+0.1	[−1.7, +1.8]
0.6875–0.8125	32.4%	30.7%	−1.7	[−4.2, +0.9]
0.875–0.9375	14.6%	13.7%	−0.9	[−2.8, +0.9]
1	16.4%	16.2%	−0.3	[−2.1, +1.8]

Mean of a respondent's four rescaled answers, which takes values in sixteenths. Sparse bins are pooled. Differences are in percentage points of respondents; intervals are 95% percentile bootstrap intervals, resampling respondents within arms.

The most direct alternative to offsetting reactions is a minority that reacts strongly against the endorser while everyone else is unmoved. That alternative also produces a small average effect and greater dispersion. To see whether it can produce the midpoint pattern, Table 3 builds endorsed arms from the control respondents themselves. In each simulated arm, a reacting respondent moves each of their four answers one step in their direction with a fixed probability, and the share of opponents is chosen so the expected average effect equals the observed one. Opposition alone, held by 5% to 22% of respondents depending on how many answers move, leaves the midpoint share where it was (changes of +0.0 to +0.1 points). Respondents leaving the midpoint downward are replaced by respondents arriving from above. In 97.5% of simulated samples of the observed size, opposition alone lowers the midpoint share by less than 1.8 points; the observed change is

−3.8 [−5.4, −2.3]. Adding a favorable group empties the midpoint: with 30% favorable and 29% opposed, the change is −2.0 points. No simulated world reproduces the observed gain one step above the midpoint (at most +0.7 points on average, against +2.7 [+1.1, +4.3] observed).

Table 3: Observed and simulated changes under endorsement

Endorsed arm	Oppose	Favor	Answers moved	Spread (×100)	Midpoint (pp)	One below (pp)	One above (pp)
Observed				+0.41 [+0.19, +0.63]	−3.8 [−5.4, −2.3]	+2.1 [+1.0, +3.1]	+2.7 [+1.1, +4.3]
Opposition only	22%	0%	25%	+0.10 [−0.11, +0.29]	+0.0 [−1.6, +1.7]	+1.0 [+0.1, +2.0]	+0.6 [−0.9, +2.0]
Opposition only	11%	0%	50%	+0.15 [−0.08, +0.35]	+0.1 [−1.6, +1.8]	+0.8 [−0.1, +1.8]	+0.4 [−1.0, +1.9]
Opposition only	5%	0%	100%	+0.26 [+0.05, +0.48]	+0.0 [−1.8, +1.9]	+0.5 [−0.4, +1.5]	+0.2 [−1.2, +1.8]
Both directions	17%	10%	50%	+0.21 [+0.01, +0.42]	−0.6 [−2.3, +1.1]	+1.2 [+0.2, +2.1]	+0.5 [−0.8, +1.9]
Both directions	23%	20%	50%	+0.28 [+0.07, +0.47]	−1.3 [−3.0, +0.3]	+1.4 [+0.5, +2.5]	+0.6 [−0.9, +2.1]
Both directions	29%	30%	50%	+0.35 [+0.13, +0.57]	−2.0 [−3.6, −0.4]	+1.7 [+0.7, +2.7]	+0.7 [−0.9, +2.3]

Changes from the control arm. The first row is the observed endorsed arm, with 95% percentile bootstrap intervals resampling respondents within arms. Other rows are simulated endorsed arms built by resampling control respondents to the size of the endorsed arm. A reacting respondent moves each answer one step (0.25) down if opposed or up if favorable, with the probability shown as answers moved, truncated at the ends of the scale. The opposed share is set so the expected average effect equals the observed one. Simulated rows report the mean and the 2.5th and 97.5th percentiles over 500 simulated samples. Spread is the variance of a respondent's mean answer. Midpoint is the share of respondents whose answers average 0.5; one below and one above are the shares averaging 0.4375 and 0.5625.

This comparison holds the counterfactual fixed at the observed control arm and varies only the direction of reactions. It assumes that whether and how a respondent reacts is unrelated to where their answers would otherwise sit. Without that assumption the marginal distributions alone guarantee only that some respondents moved down: a sharp lower bound on the share whose mean answer fell is 3.8%, while the corresponding bound for a rise is 0.2%, because a reaction that lowers only the answers of respondents just above the midpoint could refill it. An endorsement that pushed respondents toward their own policy views, without any reaction to the group, would also empty the midpoint. But it would tie answers more closely to those views and so raise the correlation between the two policies most tied to them. That correlation falls (Table 1).

These patterns do not identify a proportion of favorable reactions. They establish that endorsements produce reactions that respondents carry across questions, that those reactions are large relative to the average effect, and that where movement in both directions is observable it occurs in both directions. The small average effect reflects opposing reactions that offset, not an absence of reactions.

5 How Many React Favorably?

The within-respondent patterns show that favorable and unfavorable reactions coexist. They do not say how many respondents react favorably. Each route to that number rests on assumptions the data do not support or cannot check.

5.1 The Measurement Model

The measurement model of Bullock et al. (2011) relates ordinal policy responses to latent policy preferences and endorsement reactions. Respondent i has a latent position x_i. For question j

endorsed by group k, the model adds a reaction s_{ijk} and maps the resulting position to an observed category through question-specific loadings and cutpoints. In the baseline specification,

$$s_{ijk} \sim \mathcal{N}(\lambda_k, \omega_k^2), \qquad P(s_{ijk} > 0) = \Phi(\lambda_k/\omega_k). \tag{2}$$

The mean and variance of the reaction distribution are common across policies. The reaction itself is question-specific; the specification does not assign each respondent a single fixed supporter status across all questions. The positive probability is invariant to the units of the latent scale.

A positive latent reaction need not move a response into a higher observed category. The model also includes response noise, so $P(s_{ijk} > 0)$ is distinct from the probability that the observed treatment response exceeds the control potential response. Each reaction is drawn independently across a respondent's questions. That is the assumption Section 4 rejects: under it, the correlations between policy answers would not change.

I fit the baseline model without covariates using `endorse`.[1] Table 4 reports posterior means and credible intervals. Averaged equally over groups, the probability of a positive latent reaction is 43% (95% credible interval 32%–53%). This is an average of four marginal probabilities, not the probability of favoring at least one group. The Afghan Taliban have the lowest estimate, 32%, and a wide interval, 2%–49%. Their posterior probability of a negative mean reaction is 0.98; the other groups' mean-reaction intervals include zero.

Table 4: Positive latent reactions under the baseline measurement model

Endorser	$P(s > 0)$	95% CrI	Mean reaction (ideal-point SDs)	Reaction SD (ideal-point SDs)
Kashmiri tanzeems	45%	[25%, 58%]	−0.03	0.25
Afghan Taliban	32%	[2%, 49%]	−0.08	0.20
Al-Qaeda	47%	[40%, 54%]	−0.03	0.49
Sectarian tanzeems	49%	[38%, 57%]	−0.01	0.36
Average	43%	[32%, 53%]	−0.04	0.32

Unweighted analysis sample, $n = 5,212$. CrI denotes a 95% credible interval. Mean reaction and reaction SD are posterior means in ideal-point SD units. The average row weights the four endorsers equally.

Adding gender, rurality, education, and income covariates gives an average of 44%, and a model with division-level means gives 49%. These specifications share the independence assumption.

5.2 Bounds and Restrictions

Without a model, the experiment identifies the treatment and control distributions of each answer but not how they pair within a person. Allowing every pairing of the pooled distributions bounds the fraction of person–policy responses that an endorsement raises between 1% and 58%. The bounds are too wide to settle the question.

[1] The fitting script uses three chains with starting values dispersed across negative, zero, and positive mean reactions. Each chain runs for 200,000 iterations, discards 20,000, and retains every 40th draw. The largest reported Gelman–Rubin statistic across the mean and variance parameters is 1.016, and the smallest effective sample size across mean reactions is 1355. The fits use the package's default priors.

Fixing the sizes of favorable and unfavorable effects in Equation (1) turns the ATE into a share. Table 5 collects several such restrictions. The implied shares run from 18% to 99%, and the data contradict some of them: binary switching, for example, allows no interior responses. The response ceiling widens the range. In the control arm 42% of responses are at the top of the scale, where a favorable reaction cannot show. If favorable and unfavorable respondents both move one category, truncation at the top implies a favorable share of 59% despite a negative mean.

Table 5: Favorable shares implied by restrictions on effect sizes

Restriction	Assumption	Implied fraction
Extreme opponent reaction	$\delta^- = -1$, $\delta^+ = 0$	98.7%
Binary switching	All supporters $\to 1$, opponents $\to 0$	72.3%
5-point scale, 2-step shift	Shift ± 2 categories, truncated	65.6%
5-point scale, 1-step shift	Shift ± 1 category, truncated	58.6%
Normal heterogeneity ($\sigma = 0.05$)	$\delta_i \sim \mathcal{N}(\beta, 0.05^2)$	39.9%
Symmetric effects ($d = 0.05$)	$\delta^+ = +0.05$, $\delta^- = -0.05$	37.2%
Symmetric effects ($d = 0.02$)	$\delta^+ = +0.02$, $\delta^- = -0.02$	17.9%

All rows use the unweighted pooled ATE -0.013 and, where required, control mean 0.736. Fractions refer to the class defined by each restriction. Binary switching is contradicted by the observed interior categories. The normal row is a continuous approximation to individual effects on the observed scale; the extreme row is an unaffected fraction. Category shifts assign both classes the pooled control distribution and truncate at the scale endpoints.

5.3 Direct Polling

Pew's 2012 survey reports 14% favorable views of the Afghan Taliban, 45% unfavorable views, and 41% with no opinion (Pew Research Center, 2012). If stated opinions are classified correctly, assigning all nonrespondents to the favorable category gives an upper bound of 55%. That bound does not allow respondents who report unfavorable opinions to conceal favorable ones.

The corresponding Afghan Taliban estimate from the 2009 endorsement data is 32% (95% credible interval 2%–49%). It falls within the range obtained by varying the classification of Pew nonrespondents. But the surveys differ in year, sample composition, and measurement, and the latent probability is not the same response as direct favorability. Social desirability and fear may contribute to discrepancies; these data do not identify their contribution.

6 Discussion

The average effect of an endorsement, the fraction of responses it raises, and the prevalence of favorable attitudes are different quantities. Random assignment identifies the first. The response distributions bound the second. The third requires a model connecting responses to attitudes, and that model can be checked against how a respondent's answers move together. The standard measurement model fails that check.

A model in which a reaction shifts a respondent's answers directly, rather than through the latent position, is the natural alternative. Latent-class versions of that model, with opponents,

unaffected respondents, and supporters, did not identify the favorable share in these data: the estimate depended on how the response ceiling was modelled, the sizes of favorable and unfavorable shifts could not be separated, and simulated data with no supporters returned a positive share. The replication materials document these attempts. Population weighting and sensitivity to the assignment exclusions remain open.

Three changes to the design would identify more. Randomizing whether the group is said to support or oppose the policy separates reactions to the group from a general push away from the middle of the scale: a respondent who favors the group should move in opposite directions under the two cues, and a respondent pushed toward their own view should not. An endorsement by a neutral or widely liked actor measures how much any cue disperses answers. Policies with intermediate baseline support, or a finer response scale, leave room for favorable reactions to appear; here 42% of control responses are already at the top. Asking the policy questions without an endorsement in an earlier wave would go further, by observing each respondent's own shift and so the sizes of favorable and unfavorable reactions separately. Repeated-measures designs in survey experiments have generally not altered treatment effects (Clifford et al., 2021), although consistency pressure across waves would need to be checked.

7 Conclusion

A small negative average endorsement effect does not mean respondents are indifferent to militant groups. In these data it is the net of reactions in both directions: answers to most policy pairs become more closely related under endorsement, and the middle of the scale empties in a way that opposition alone does not produce. How many respondents react favorably is not identified. The fitted measurement model puts the probability of a positive latent reaction at 43%, but under an assumption the data reject. Moving from an average response effect to a distribution of attitudes requires stating and defending the additional assumptions, and designs that flip the direction of the cue or observe each respondent's own shift would require fewer of them.

References

Blair, G., Fair, C. C., Malhotra, N., and Shapiro, J. N. (2013). Poverty and support for militant politics: Evidence from Pakistan. *American Journal of Political Science*, 57(1):30–48.

Blair, G., Imai, K., and Lyall, J. (2014). Comparing and combining list and endorsement experiments: Evidence from Afghanistan. *American Journal of Political Science*, 58(4):1043–1063.

Bullock, W., Imai, K., and Shapiro, J. N. (2011). Statistical analysis of endorsement experiments: Measuring support for militant groups in Pakistan. *Political Analysis*, 19(4):363–384.

Clifford, S., Sheagley, G., and Piston, S. (2021). Increasing precision without altering treatment effects: Repeated measures designs in survey experiments. *American Political Science Review*, 115(3):1048–1065.

Lyall, J., Blair, G., and Imai, K. (2013). Explaining support for combatants during wartime: A survey experiment in Afghanistan. *American Political Science Review*, 107(4):679–705.

Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review*, 80(2):319–323.

Pew Research Center (2012). Pakistani public opinion ever more critical of U.S. Technical report, Pew Research Center.

Shiraito, Y. and Imai, K. (2022). *endorse: Bayesian Measurement Models for Analyzing Endorsement Experiments*. R package version 1.6.2.